Assignment no 18

Code

Student.sol

```
// SPDX-License-Identifier: UNLICENSED
pragma solidity ^0.8.0;

contract StudentData {

    // Structure
    struct Student {
        string name;
        uint256 rollno;
    }

    // Array
    Student[] public studentarr;

    function addStudent(string memory name, uint rollno) public {
        for (uint i = 0; i < studentarr.length; i++) {
            if (studentarr[i].rollno == rollno) {
                revert("Student with this roll no already exists");
            }
        }

        studentarr.push(Student(name, rollno));
    }

    function getStudentsLength() public view returns (uint) {
        return studentarr.length;
    }

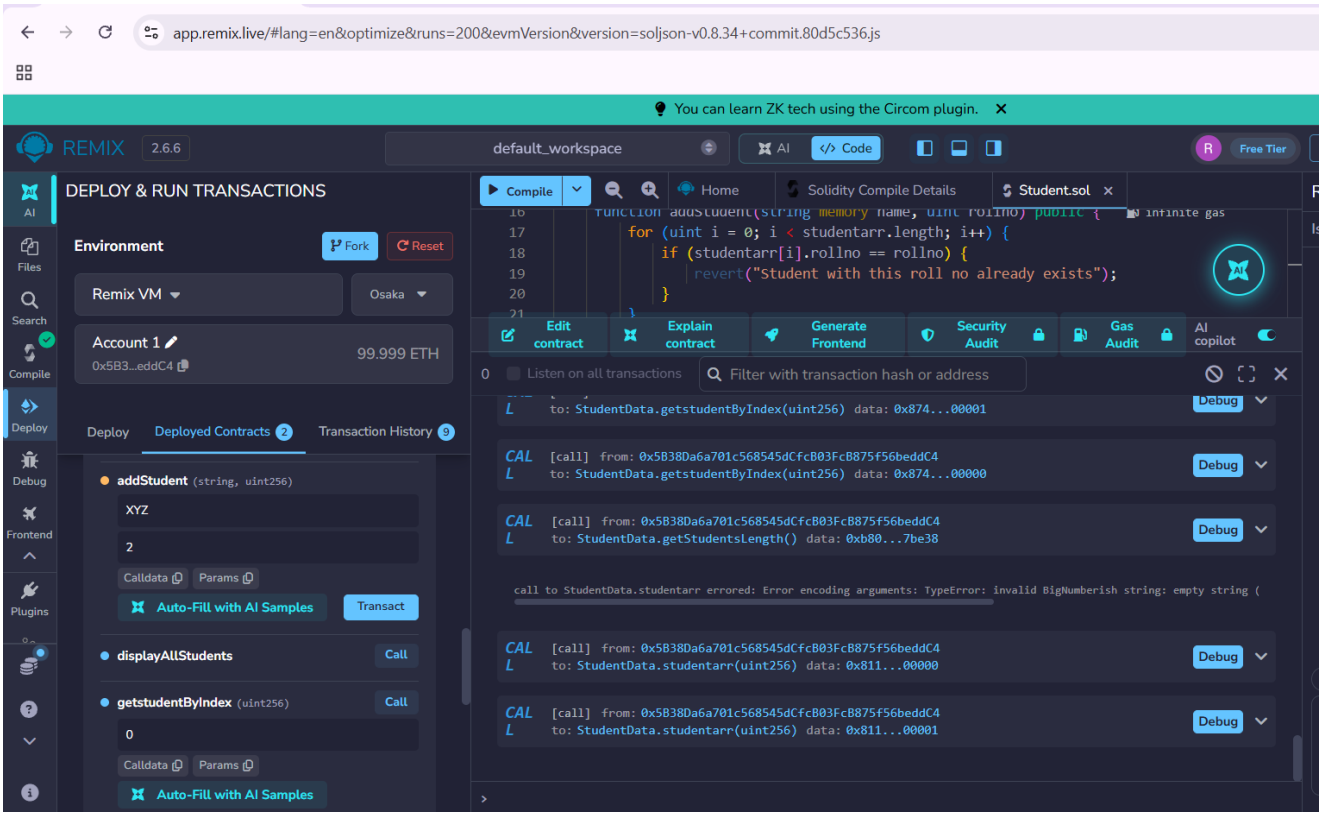
    function displayAllStudents() public view returns (Student[] memory) {
        return studentarr;
    }

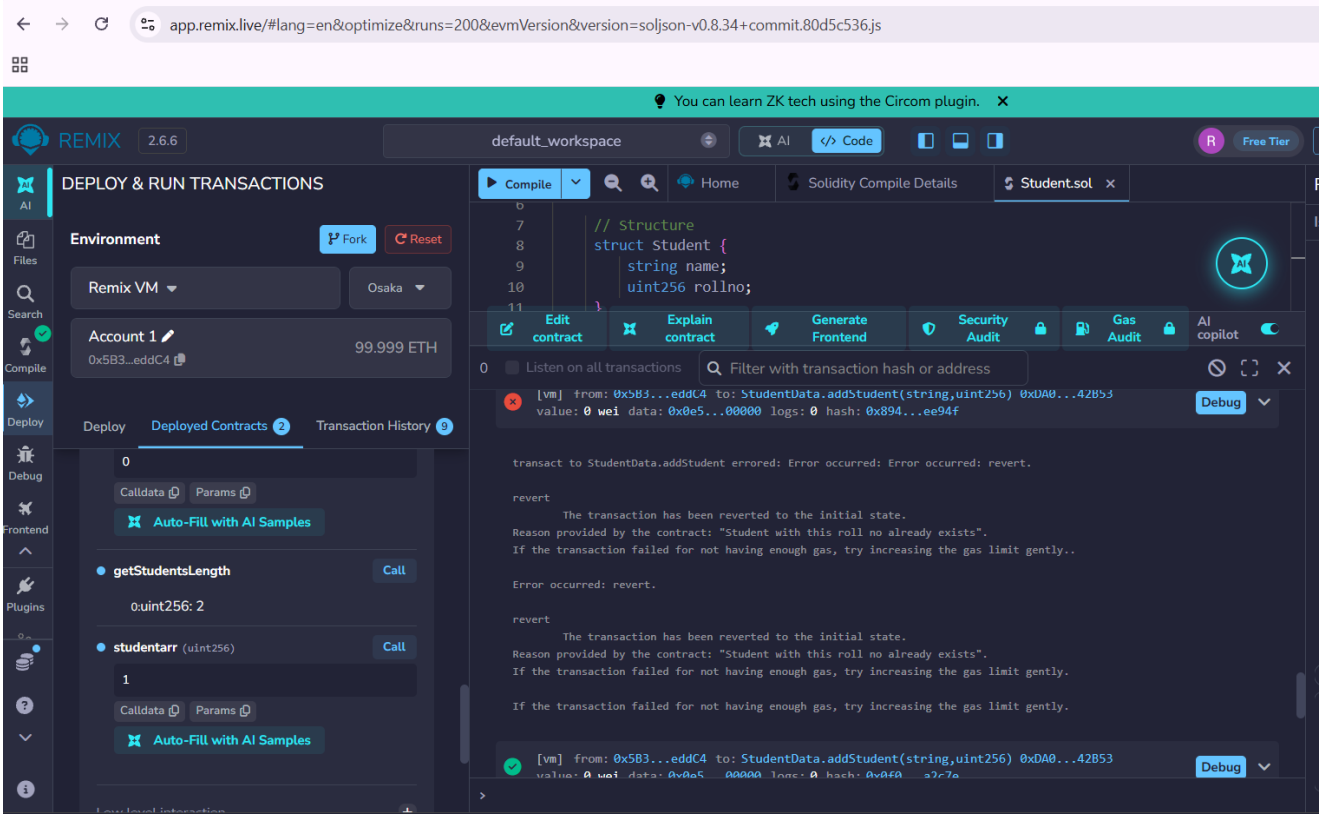
    function getstudentByIndex(uint idx) public view returns (Student memory) {
        require(studentarr.length > idx, "index out of bound");
        return studentarr[idx];
    }

    // Fallback
    fallback() external payable {
        // This function will handle an external function call
        // that is not there in our contract.
    }

    receive() external payable {
        // This function will handle the Ether sent by an external user
        // without data mentioned.
    }
}
```

Output


app.remix.live/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js
You can learn ZK tech using the Circom plugin.
REMIX 2.6.6
default_workspace
DEPLOY & RUN TRANSACTIONS
Environment
Fork
Reset
Remix VM
Osaka
Account 1
0x5B3...eddC4
99.999 ETH
Deploy
Deployed Contracts 2
Transaction History 9
addStudent (string, uint256)
XYZ
2
Calldata
Params
Auto-Fill with AI Samples
Transact
displayAllStudents
Call
getstudentByIndex (uint256)
0
Compile
Home
Solidity Compile Details
Student.sol
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
to: StudentData.getstudentByIndex(uint256) data: 0x874...00001
[call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
to: StudentData.getstudentByIndex(uint256) data: 0x874...00000
to: StudentData.getStudentsLength() data: 0xb80...7be38
call to StudentData.studentarr errored: Error encoding arguments: TypeError: invalid BigNumberish string: empty string (
to: StudentData.studentarr(uint256) data: 0x811...00000
to: StudentData.studentarr(uint256) data: 0x811...00001
Debug


app.remix.live/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js
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0:uint256: 2
studentarr (uint256)
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// Structure
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Edit contract
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Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
[vm] from: 0x5B3...eddC4 to: StudentData.addStudent(string,uint256) 0xDA0...42B53
value: 0 wei data: 0x0e5...00000 logs: 0 hash: 0x894...ee94f
transact to StudentData.addStudent errored: Error occurred: Error occurred: revert.
revert
The transaction has been reverted to the initial state.
Reason provided by the contract: "Student with this roll no already exists".
If the transaction failed for not having enough gas, try increasing the gas limit gently..
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Debug